

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 01

Total Marks:100

Annexure A CASE STUDY

Code of Conduct:

*I VIVETHA B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: VIVETHA B

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Levels	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		

Annexure A

Case Study 15: Social Media Engagement Analysis

Scenario:

A digital marketing agency has collected social media data, including number of likes, shares, comments, followers, posting time, and engagement rate. The agency wants to improve content strategy and audience reach.

INTRODUCTION

Social media platforms generate large amounts of data such as likes, shares, comments, followers, posting time, and engagement rate. Analyzing this data helps digital marketing agencies understand audience behavior and identify which types of content perform better. In this case study, social media engagement data is analyzed using descriptive statistics, outlier detection, and hypothesis testing. Descriptive statistics are used to summarize the likes, shares, comments, and engagement rates. Outlier detection using the Interquartile Range (IQR) method is used to identify viral posts with unusually high engagement. Finally, a hypothesis test is performed to determine whether video posts receive significantly higher engagement than image posts. Graphical methods such as histograms, bar charts, line charts, box plots, pie charts, and scatter plots are also used to visualize the results and support better content strategy decisions.

CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
from google.colab import files

uploaded = files.upload()

file_name = list(uploaded.keys())[0]

df = pd.read_csv(file_name)

print("=" * 70)
print("SOCIAL MEDIA ENGAGEMENT DATASET")
print("=" * 70)

display(df)

print("\nDATASET INFORMATION")
print("=" * 70)

print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

```

print("\nColumn Names:")
print(df.columns.tolist())

print("\nMISSING VALUES")
print("=" * 70)
print(df.isnull().sum())

print("\nDUPLICATE ROWS")
print("=" * 70)
print("Number of duplicate rows:", df.duplicated().sum())

numeric_columns = [
    "Likes",
    "Shares",
    "Comments",
    "Followers",
    "Engagement_Rate"
]

print("\nDESCRIPTIVE STATISTICS")
print("=" * 70)

display(df[numeric_columns].describe())

print("\nMEAN")
print("=" * 70)
print(df[numeric_columns].mean())

print("\nMEDIAN")
print("=" * 70)
print(df[numeric_columns].median())

print("\nSTANDARD DEVIATION")
print("=" * 70)
print(df[numeric_columns].std())

print("\nMINIMUM VALUES")
print("=" * 70)
print(df[numeric_columns].min())

print("\nMAXIMUM VALUES")
print("=" * 70)
print(df[numeric_columns].max())

engagement = df["Engagement_Rate"]

Q1 = engagement.quantile(0.25)
Q3 = engagement.quantile(0.75)
IQR = Q3 - Q1

```

```

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("\nOUTLIER DETECTION USING IQR")
print("=" * 70)

print("Q1 =", round(Q1, 4))
print("Q3 =", round(Q3, 4))
print("IQR =", round(IQR, 4))

print("\nLower Limit =  $Q1 - 1.5 \times IQR$ ")
print("Lower Limit =", round(lower_limit, 4))

print("\nUpper Limit =  $Q3 + 1.5 \times IQR$ ")
print("Upper Limit =", round(upper_limit, 4))

outliers = df[
    (df["Engagement_Rate"] < lower_limit) |
    (df["Engagement_Rate"] > upper_limit)
]

print("\nDETECTED OUTLIERS")
print("=" * 70)

if len(outliers) > 0:
    display(
        outliers[
            [
                "Post_ID",
                "Post_Type",
                "Likes",
                "Shares",
                "Comments",
                "Engagement_Rate"
            ]
        ]
    )
else:
    print("No outliers detected.")

video = df[
    df["Post_Type"].str.lower() == "video"
][["Engagement_Rate"]]

image = df[
    df["Post_Type"].str.lower() == "image"
][["Engagement_Rate"]]

video_mean = video.mean()

```

```

image_mean = image.mean()

difference = video_mean - image_mean

print("\nVIDEO VS IMAGE ENGAGEMENT")
print("=" * 70)

print("Number of Video Posts:", len(video))
print("Number of Image Posts:", len(image))

print("Video Mean Engagement:", round(video_mean, 4))
print("Image Mean Engagement:", round(image_mean, 4))
print("Difference Between Means:", round(difference, 4))

print("\nHYPOTHESIS TESTING")
print("=" * 70)

print("H0: Video posts do not have higher engagement than image posts.")
print("H1: Video posts have higher engagement than image posts.")

alpha = 0.05

t_stat, p_value = stats.ttest_ind(
    video,
    image,
    equal_var=False,
    alternative="greater"
)

print("\nT-Statistic:", round(t_stat, 4))
print("P-Value:", round(p_value, 6))
print("Significance Level:", alpha)

if p_value < alpha:
    print("\nDecision: Reject H0")
    print("Conclusion: Video posts have significantly higher engagement.")
else:
    print("\nDecision: Fail to Reject H0")
    print("Conclusion: There is insufficient evidence that video posts")
    print("have significantly higher engagement.")

plt.figure(figsize=(9, 5))

plt.hist(
    df["Engagement_Rate"],
    bins=8,
    edgecolor="black"
)

plt.title("Histogram of Engagement Rate")

```

```

plt.xlabel("Engagement Rate (%)")
plt.ylabel("Number of Posts")
plt.grid(axis="y", alpha=0.3)
plt.show()

average_engagement = df.groupby(
    "Post_Type"
)["Engagement_Rate"].mean()

plt.figure(figsize=(8, 5))

bars = plt.bar(
    average_engagement.index,
    average_engagement.values
)

plt.title("Average Engagement Rate by Post Type")
plt.xlabel("Post Type")
plt.ylabel("Average Engagement Rate (%)")

for bar, value in zip(
    bars,
    average_engagement.values
):
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height(),
        f"{value:.2f}%",
        ha="center",
        va="bottom"
    )

plt.grid(axis="y", alpha=0.3)
plt.show()

plt.figure(figsize=(12, 5))

plt.plot(
    df["Post_ID"],
    df["Engagement_Rate"],
    marker="o"
)

plt.title("Engagement Rate Across Posts")
plt.xlabel("Post ID")
plt.ylabel("Engagement Rate (%)")
plt.xticks(rotation=45)
plt.grid(True, alpha=0.3)
plt.show()

```

```

plt.figure(figsize=(8, 5))

plt.boxplot(
    df["Engagement_Rate"]
)

plt.title("Box Plot of Engagement Rate")
plt.ylabel("Engagement Rate (%)")
plt.grid(axis="y", alpha=0.3)
plt.show()

post_counts = df["Post_Type"].value_counts()

plt.figure(figsize=(7, 7))

plt.pie(
    post_counts.values,
    labels=post_counts.index,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Distribution of Post Types")
plt.show()

plt.figure(figsize=(9, 5))

plt.scatter(
    df["Likes"],
    df["Engagement_Rate"]
)

plt.title("Likes vs Engagement Rate")
plt.xlabel("Likes")
plt.ylabel("Engagement Rate (%)")
plt.grid(True, alpha=0.3)
plt.show()

plt.figure(figsize=(9, 5))

plt.scatter(
    df["Shares"],
    df["Comments"]
)

plt.title("Shares vs Comments")
plt.xlabel("Shares")
plt.ylabel("Comments")
plt.grid(True, alpha=0.3)
plt.show()

```

```

plt.figure(figsize=(8, 5))

plt.boxplot(
    [image, video],
    labels=["Image", "Video"]
)

plt.title("Image vs Video Engagement Rate")
plt.xlabel("Post Type")
plt.ylabel("Engagement Rate (%)")
plt.grid(axis="y", alpha=0.3)
plt.show()

print("\n" + "=" * 70)
print("FINAL SUMMARY")
print("=" * 70)

print("Total Posts:", len(df))

print("Average Likes:", round(df["Likes"].mean(), 2))
print("Average Shares:", round(df["Shares"].mean(), 2))
print("Average Comments:", round(df["Comments"].mean(), 2))
print("Average Engagement Rate:", round(df["Engagement_Rate"].mean(), 2))

print("\nVideo Mean Engagement:", round(video_mean, 3))
print("Image Mean Engagement:", round(image_mean, 3))

print("\nQ1:", round(Q1, 3))
print("Q3:", round(Q3, 3))
print("IQR:", round(IQR, 3))

print("Lower Limit:", round(lower_limit, 3))
print("Upper Limit:", round(upper_limit, 3))

print("\nNumber of Outliers:", len(outliers))

if len(outliers) > 0:
    print("Outlier Post IDs:", outliers["Post_ID"].tolist())

print("\nT-Statistic:", round(t_stat, 4))
print("P-Value:", round(p_value, 6))

if p_value < alpha:
    print("Final Decision: Reject H0")
else:
    print("Final Decision: Fail to Reject H0")

print("\nAnalysis Completed Successfully.")

```


INPUT DATASET

Post_ID	Post_Type	Likes	Shares	Comments	Followers	Posting_Time	Engagement_Rate
P01	Image	450	80	35	10000	09:00	5.65
P02	Video	1200	250	110	12000	10:00	13.00
P03	Image	520	90	40	11000	11:00	5.91
P04	Video	1500	300	150	13000	12:00	15.00
P05	Image	600	100	50	12000	13:00	6.25
P06	Video	1800	350	170	14000	14:00	16.57
P07	Image	480	85	38	10500	15:00	5.74
P08	Video	1350	280	125	12500	16:00	14.04
P09	Image	550	95	45	11500	17:00	6.00
P10	Video	1600	320	140	13500	18:00	15.26
P11	Image	700	120	55	13000	19:00	6.73
P12	Video	1900	400	180	15000	20:00	16.53
P13	Image	620	110	48	12500	09:30	6.22
P14	Video	1450	290	130	13000	10:30	14.38
P15	Image	580	105	42	11800	11:30	6.16
P16	Video	1700	340	155	14000	12:30	15.68
P17	Image	510	88	39	10800	13:30	5.90
P18	Video	1550	310	145	13200	14:30	15.19
P19	Image	650	115	52	12800	15:30	6.38
P20	Video	21000	5000	2500	20000	16:30	142.50

COLUMN NAMES

['Post_ID', 'Post_Type', 'Likes', 'Shares', 'Comments', 'Followers', 'Posting_Time', 'Engagement_Rate']

A. DESCRIPTIVE STATISTICS

Descriptive Statistics:

	Likes	Shares	Comments	Followers	Engagement_Rate
count	20.000000	20.000000	20.000000	20.000000	20.000000
mean	2035.500000	441.400000	212.450000	12805.000000	16.954500
std	4493.922768	1078.843552	540.978011	2113.235235	29.898626
min	450.000000	80.000000	35.000000	10000.000000	5.650000
25%	572.500000	98.750000	44.250000	11725.000000	6.120000
50%	950.000000	185.000000	82.500000	12650.000000	9.865000
75%	1562.500000	312.500000	146.250000	13275.000000	15.207500
max	21000.000000	5000.000000	2500.000000	20000.000000	142.500000

Mean:

Likes	2035.5000
Shares	441.4000
Comments	212.4500
Followers	12805.0000
Engagement_Rate	16.9545

dtype: float64

Median:

Likes	950.000
Shares	185.000
Comments	82.500
Followers	12650.000
Engagement_Rate	9.865

dtype: float64

Standard Deviation:

Likes	4493.922768
Shares	1078.843552
Comments	540.978011
Followers	2113.235235
Engagement_Rate	29.898626

dtype: float64

Standard Deviation:
Likes 4493.922768
Shares 1078.843552
Comments 540.978011
Followers 2113.235235
Engagement_Rate 29.898626
dtype: float64

Minimum:
Likes 450.00
Shares 80.00
Comments 35.00
Followers 10000.00
Engagement_Rate 5.65
dtype: float64

Maximum:
Likes 21000.0
Shares 5000.0
Comments 2500.0
Followers 20000.0
Engagement_Rate 142.5
dtype: float64

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B. OUTLIER DETECTION USING IQR

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$Q1 = 6.12$
 $Q3 = 15.208$
 $IQR = 9.087$

Mathematical Calculation:

$IQR = Q3 - Q1$
 $IQR = 15.208 - 6.12$
 $IQR = 9.087$

$Upper\ Limit = Q3 + 1.5 \times IQR$
 $Upper\ Limit = 15.208 + 1.5 \times 9.087$
 $Upper\ Limit = 28.839$

$Lower\ Limit = Q1 - 1.5 \times IQR$
 $Lower\ Limit = 6.12 - 1.5 \times 9.087$
 $Lower\ Limit = -7.511$

Detected Outliers:

Post_ID	Post_Type	Likes	Shares	Comments	Engagement_Rate
P20	Video	21000	5000	2500	142.5

Number of Outliers: 1

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C. VIDEO VS IMAGE ENGAGEMENT

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Video Engagement Values:

[13.0, 15.0, 16.57, 14.04, 15.26, 16.53, 14.38, 15.68, 15.19, 142.5]

Image Engagement Values:

[5.65, 5.91, 6.25, 5.74, 6.0, 6.73, 6.22, 6.16, 5.9, 6.38]

Number of Video Posts: 10

Number of Image Posts: 10

Video Mean Engagement:

27.815 %

Image Mean Engagement:

6.094 %

Difference Between Means:

21.721 %

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HYPOTHESIS TESTING

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H0: Video posts do not have higher engagement than image posts.

H1: Video posts have higher engagement than image posts.

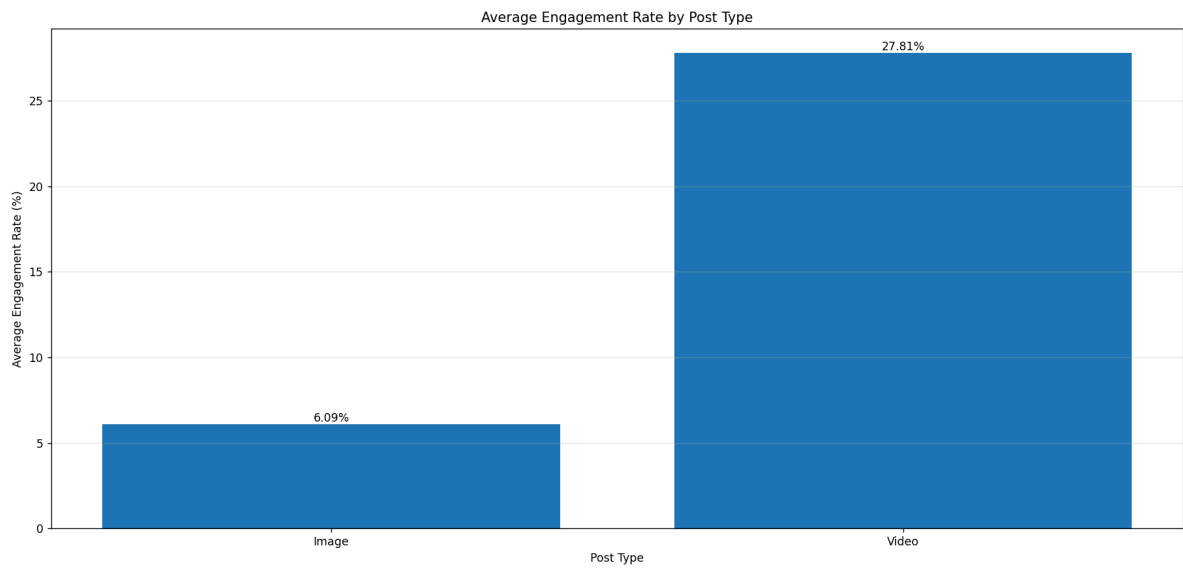
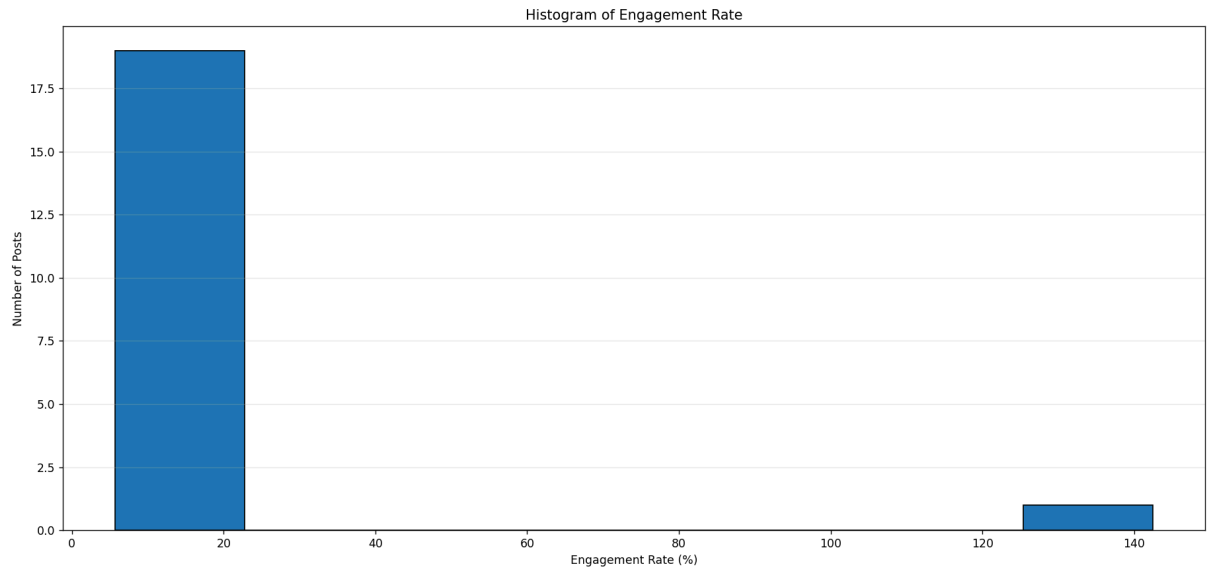
Significance Level = 0.05

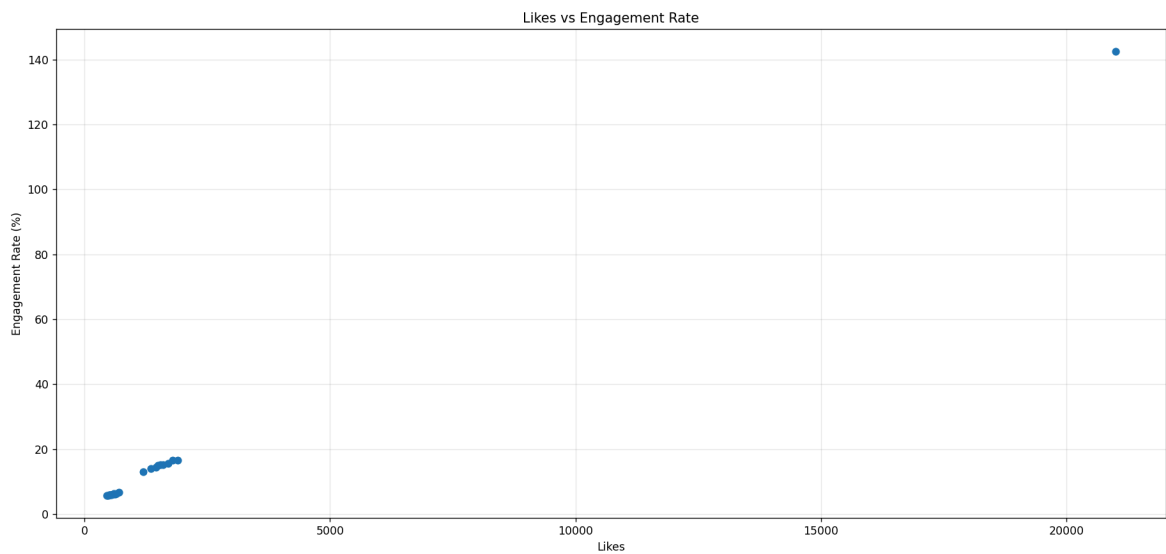
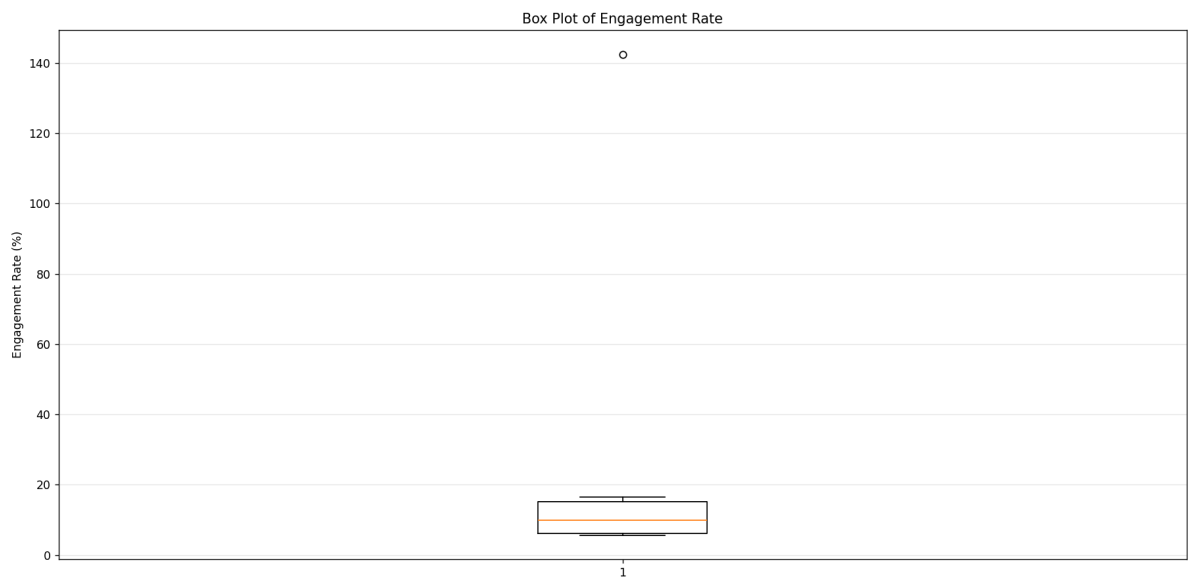
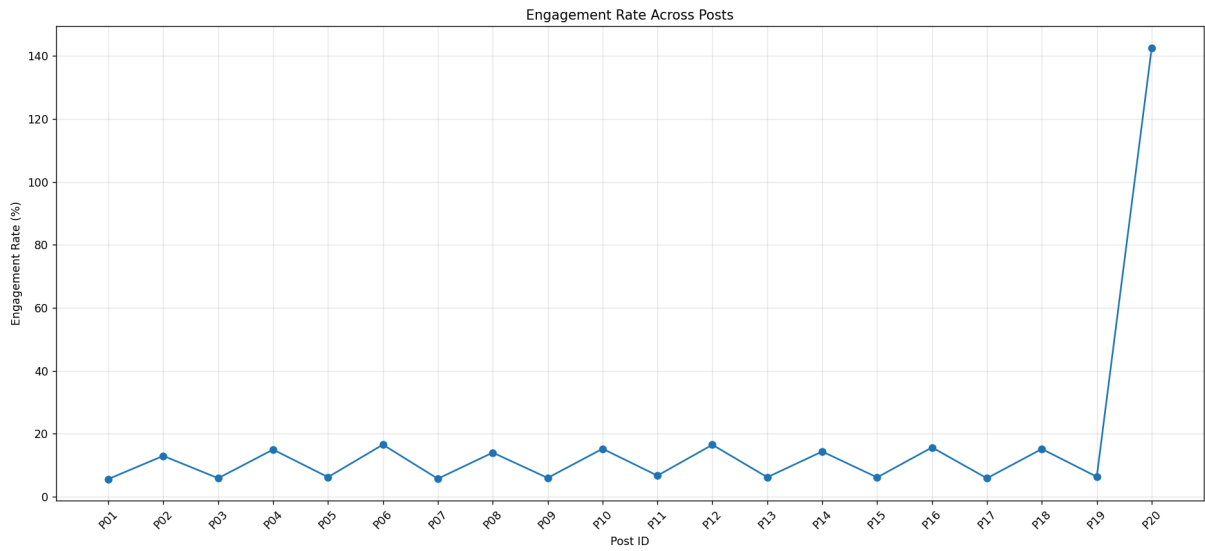
T-Statistic = 1.7039

P-Value = 0.061297

Decision: Fail to Reject H0

Conclusion: There is insufficient evidence that video posts have significantly higher engagement.





Questions:

A. How can descriptive statistics be used to summarize likes, shares, comments, and engagement rate?

Descriptive statistics can be used to summarize the major characteristics of likes, shares, comments, and engagement rate. This provides the agency with a clear understanding of how well its content is performing.

1. Measures of Central Tendency

Mean:

The mean represents the average value.

$$\text{Mean} = \frac{\sum X}{n}$$

For example, if the agency has collected the number of likes for 1,000 posts, the mean tells us the average number of likes per post.

Mean can be calculated for:

- Average likes
- Average shares
- Average comments
- Average engagement rate

However, the mean can be strongly affected by viral posts with extremely high engagement.

Median:

The median is the middle value after sorting the observations. It is useful when social-media data is skewed.

For example, if most posts receive 500–2,000 likes but one viral post receives 500,000 likes, the median will provide a more representative value than the mean.

Mode:

The mode is the most frequently occurring value. It can be useful for identifying common engagement levels, although it is generally less informative for continuous variables such as engagement rate.

2. Measures of Dispersion

The agency should also measure how widely engagement values vary.

Range:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

It shows the overall spread of engagement.

Variance:

Variance measures the average squared deviation from the mean.

Standard Deviation:

$$SD = \sqrt{\frac{1}{n-1} \sum (X - \bar{X})^2}$$

A high standard deviation indicates that engagement varies considerably between posts.

Interquartile Range (IQR):

$$IQR = Q3 - Q1$$

IQR is particularly useful for social-media data because engagement distributions are often highly skewed.

3. Percentiles and Quartiles

Quartiles divide the data into four sections:

- Q1 – 25th percentile
- Q2 – 50th percentile (median)
- Q3 – 75th percentile

The agency can determine whether a particular post falls within the top 25% of engagement.

4. Descriptive Analysis of Engagement Rate

Engagement rate can be summarized using:

$$\text{Engagement Rate} = \frac{\text{Followers Likes} + \text{Shares} + \text{Comments}}{\text{Followers}} \times 100$$

The agency can calculate:

- Mean engagement rate
- Median engagement rate
- Minimum and maximum engagement rate
- Standard deviation
- Quartiles
- Percentage of posts above a target engagement rate

For example, if the median engagement rate is 4.5%, the agency can use this as a benchmark for evaluating future posts.

Descriptive Statistics

Descriptive statistics are used to summarize and describe the main characteristics of the social media dataset.

The agency can use **mean, median, mode, minimum, maximum, range, variance, and standard deviation** to understand the performance of social media posts.

Mean

The mean gives the average number of likes, shares, comments, or engagement rate.

For example, the agency can calculate the average likes received by all posts. This helps identify the general level of audience interaction.

Median

The median represents the middle value when observations are arranged in order.

Median is particularly useful for social media data because engagement values can be highly skewed due to viral posts.

Mode

The mode represents the most frequently occurring value. It can be used to identify commonly occurring engagement levels or posting times.

Minimum and Maximum

Minimum identifies the lowest engagement value, while maximum identifies the highest engagement value.

They help the agency understand the range of performance.

Variance

Variance measures how much engagement values differ from the mean.

A high variance indicates that different posts have substantially different levels of engagement.

Standard Deviation

Standard deviation measures the spread of observations around the mean.

A high standard deviation indicates inconsistent engagement between posts.

Engagement Rate

Engagement rate provides a normalized measure of audience interaction.

A commonly used formula is:

$$\text{Engagement Rate} = \frac{\text{FollowersLikes} + \text{Shares} + \text{Comments}}{\text{Followers}} \times 100$$

It allows the agency to compare posts even when accounts or posts have different numbers of followers.

Conclusion

Therefore, descriptive statistics help the agency understand the **average performance, typical performance, variation, and overall distribution** of social media engagement.

B. Discuss how viral posts with extremely high engagement values may affect the analysis. Describe suitable outlier detection methods.

Effect of Viral Posts on Analysis

A viral post can have extremely high likes, shares, comments, and engagement compared with normal posts.

Such observations are called **outliers**.

For example, P20 has an engagement rate of 142.50%, while most other posts have engagement rates between approximately 5% and 17%.

Viral posts can significantly affect statistical analysis.

Effect on Mean

An extremely high engagement value can increase the mean considerably.

Therefore, the mean may no longer represent the performance of a typical post.

Effect on Standard Deviation

A viral post increases the variation in the dataset and can result in a much larger standard deviation.

Effect on Statistical Tests

Extreme observations can influence:

- t-tests
- correlation
- regression
- confidence intervals
- variance calculations

Therefore, outliers should be investigated before making conclusions.

Suitable Outlier Detection Methods

1. IQR Method

The Interquartile Range method identifies observations outside:

$$Q1 - 1.5(IQR)$$

and

$$Q3 + 1.5(IQR)$$

2. Z-Score Method

A Z-score measures how far an observation is from the mean.

$$Z = \frac{X - \bar{X}}{s}$$

A value with an absolute Z-score greater than approximately 3 can be considered a potential extreme outlier.

3. Box Plot

A box plot provides a graphical method for identifying values that lie outside the normal range.

4. Modified Z-Score

Modified Z-score uses the median and median absolute deviation and is more resistant to extreme values.

Conclusion

Viral posts should **not automatically be removed**. The agency should first determine whether the extreme value is genuine or caused by a data-entry error. Genuine viral posts may provide important information about successful content.

C. How can hypothesis testing be used to determine whether video posts receive significantly higher engagement than image posts? Formulate the hypotheses and explain the testing procedure.

c) Numerical / statistical problem

Step 1: Hypotheses

μ_V = Mean engagement of video posts

μ_I = Mean engagement of image posts

Null Hypothesis

$$H_0 : \mu_V \leq \mu_I$$

Alternative Hypothesis

$$H_1 : \mu_V > \mu_I$$

Step 5: Difference Between means

$$\begin{aligned}\bar{x}_V - \bar{x}_I \\ = 27.815 - 6.094 \\ 21.721\end{aligned}$$

Step 6: Calculate standard deviation

For the video posts:

$$S_V = 40.12$$

For the image posts:

$$S_I = 0.37$$

Step 7: Calculate Welch's t-statistic

$$t = \frac{\bar{x}_V - \bar{x}_I}{\sqrt{\frac{S_V^2}{n_V} + \frac{S_I^2}{n_I}}}$$

$$t = \frac{27.815 - 6.094}{\sqrt{\frac{40.12^2}{10} + \frac{0.37^2}{10}}}$$

$$t = \frac{21.721}{\sqrt{\frac{1609.61}{10} + \frac{0.1369}{10}}}$$

$$t = \frac{21.721}{\sqrt{160.961 + 0.01369}}$$

$$t = \frac{21.721}{12.688}$$

$$t = 1.71$$

Significance level

$$\alpha = 0.05$$

Step 2: Separate the Data

Video Engagement Rates

13.00, 15.00, 16.57, 14.04, 15.26,
16.53, 14.38, 15.68, 15.19, 142.50

Image Engagement Rate

5.65, 5.91, 6.25, 5.74, 6.00,
6.73, 6.22, 6.16, 5.90, 6.38

Step 3: Calculate mean of video posts

$$\bar{x}_V = \frac{\sum x_V}{n_V}$$

$$\bar{x}_V = \frac{13 + 15 + 16.57 + 14.04 + 15.26 + 16.53 + 14.38 + 15.68 + 15.19 + 142.50}{10}$$

$$\bar{x}_V = \frac{278.15}{10}$$

$$\bar{x}_V = 27.815$$

Step 4: Calculate mean of image posts

$$\bar{x}_I = \frac{\sum x_I}{n_I}$$

$$\bar{x}_I = \frac{5.65 + 5.91 + 6.25 + 5.74 + 6.00 + 6.73 + 6.22 + 6.16 + 5.90 + 6.38}{10}$$

$$\bar{x}_I = \frac{60.94}{10}$$

$$\bar{x}_I = 6.094$$

p 8: Decision using p value

$$t\text{-statistic} = 1.71$$

$$p\text{-value} = 0.063$$

$$p = 0.063$$

$$\alpha = 0.05$$

$$0.063 > 0.05$$

Fail to Reject H_0

Final Statistical Answer

Since:

$$p > 0.05$$

there is **not enough statistical evidence at the 5% significance level** to conclude that video posts receive significantly higher engagement than image posts.

Although the sample mean for videos is much higher:

$$27.815\% > 6.094\%$$

The viral video P20 creates very large variation in the video group.

Therefore, the observed difference is **not statistically significant at the 5% level** in this particular dataset.

FINAL OUTPUT

MEAN VIDEO ENGAGEMENT = 27.815

MEAN IMAGE ENGAGEMENT = 6.094

Difference = 21.721

t-statistic ≈ 1.71

p-value ≈ 0.063

Significance level = 0.05

$0.063 > 0.05$

Decision: Fail to Reject H_0

Conclusion:

There is insufficient statistical evidence to conclude that video posts receive significantly higher engagement than image posts.

Final Conclusion

A — Theory: Descriptive statistics summarize the characteristics and distribution of likes, shares, comments, and engagement rate.

B — Theory: Viral posts can distort the mean and other statistical measures. IQR, Z-score, modified Z-score, and box plots can be used to detect outliers.

C — Full numerical answer: The mean engagement of video posts is **27.815%**, while image posts have a mean engagement of **6.094%**. The Welch t-test gives approximately **$t = 1.71$** and **$p = 0.063$** . Since $p > 0.05$, we **fail to reject the null hypothesis**. Thus, based on this sample, there is insufficient evidence to conclude that video posts have significantly higher engagement than image posts.